

Paramétrage :
- HVLP V/F setup

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V/F Control Overview

- Induction machine (with or without an encoder)
- Open loop frequency based control with current limit
- Easy to set up – no dyno required, inherent field weakening
- Efficiency worse than flux vector
- Not to be used for traction
- Always in speed mode (but no speed gains)
- Still under development so liable to change

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V/F Control Operation

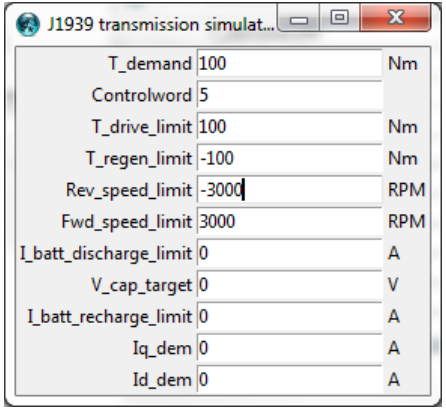
- Control still works in DQ vectors (but not really)
  - Small **Ud voltage** is applied to give rated magnetising current at stall
  - Uq is applied proportionally to frequency (from speed demand)
- Feedforward terms:
  - Ud = Rs * Im_rated * Rs_compensation_factor
  - Uq = lm_rated * Lm_rated * frequency_demand

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V/F Control Interface

- Interface same as other motor algorithms (H-protocol / J1939)
  - Torque demand (used as a switch to determine if the forward or reverse speed limit is used as the demand), also clamps the torque limits
  - Drive and regen torque limits
  - Forward and reverse speed limits



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V/F Control Interface

- Inputs
  - Torque limits directly converted to current limit
    - Set **max torque** 0x6076 to 100Nm to treat it as a percentage demand
  - Speed limits/demand directly converted to frequency
    - Will calculate from pole pairs
- Outputs
  - **Applied frequency** is converted to a **speed in RPM** in 0x606C, 0
  - If an encoder is used the speed is in 0x4631, 14
  - Torque actual 0x6077, 0 is calculated from percentage of Iq current

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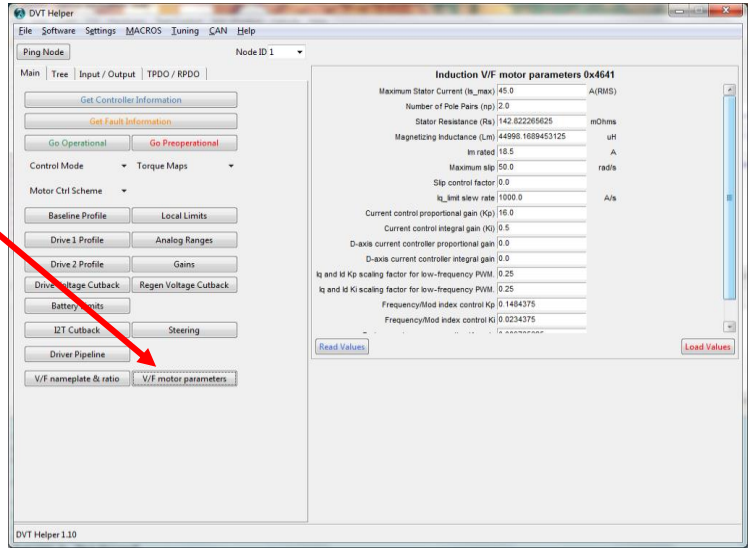
V/F Control Parameters

- Control gains – 3 control loops
  - **Rs compensation** when enabling (1kHz)
    - Runs for 500ms (although going to be changed to be more adaptive)
    - Kp 0x4641, 48 and Ki 0x4641, 49
  - **Current limit** using applied frequency (4kHz) – might be increased to 16kHz
    - Limit is calculated from iq_limit = sqrt(is_max^2 – im_rated^2)
    - Kp 0x4641, 25 and Ki 0x4641, 26
  - **Current limit** using voltage (16kHz) - likely to be removed
    - Controls stator current (Is) using Uq voltage
    - Limit is fixed at power frame limit 48A (air cooled) and 53A (water cooled)
    - Kp 0x4641, 13 and Ki 0x4641, 15
    - Leave D axis gains at zero

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V/F Control Parameters

- Key motor parameters:
  - Is_max
  - pole pairs
  - Rs_phase
  - Lm_rated
  - Im_rated
  - Max slip
  - Slip control factor
- Tables
  - Saturation map, flux map etc not used
  - Power limit map (operates in speeds in rpm and torques in Nm) is used but can be disabled by setting the torque values all to max as not required



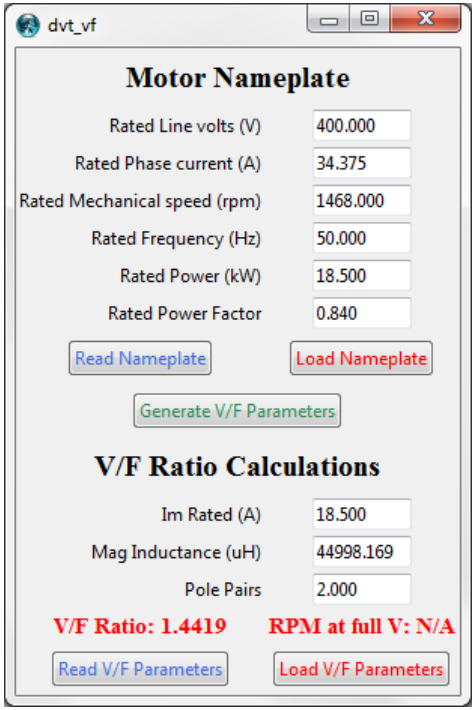
V/F Control Setup

- Setup procedure
  - Decide whether to wire the motor star or delta (if available)
  - Measure motor stator resistance between two terminals, typical resistances are in ohms magnitude. Divide by 2 to give phase resistance **(always assuming star)**
  - Measured resistance will typically not give correct Im at power up due to incorrect deadtime compensation, but don't change Rs as PI will compensate for this – you may need to tune the Rs compensation gains

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V/F Control Setup

- Setup procedure
  - DVT helper V/F window
    - Enter nameplate
      - *(cliquer sur / click on →"Load Nameplete")*
      - *(cliquer sur / click on →"Generate V/F Parameters")*
    - Will calculate:
      - poles pairs
      - Im_rated
      - Lm_rated
      - *(cliquer sur / click on →"Load V/F Parameters")*



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V/F Control Setup

- Setup procedure
  - Run motor at speed below field weakening (preferably no load) and tune Lm to give rated magnetising current in Id (typically need to increase Lm)
  - Tune frequency current limit gains if required, higher gains are required for faster acceleration, or reduced to limit oscillations
  - Can usually leave voltage current limit gains at defaults
  - Configurable speed and torque ramp rates:
    - Speed accel/decel rate : 0x6083
      - Sub 1 acceleration rate
      - Sub 2 deceleration rate
      - Sub 3 rate of change of acceleration (s-curve)
    - Torque slope : 0x6087
      - Sub 1 increase torque rate
      - Sub 2 decrease torque rate

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V/F Control Setup

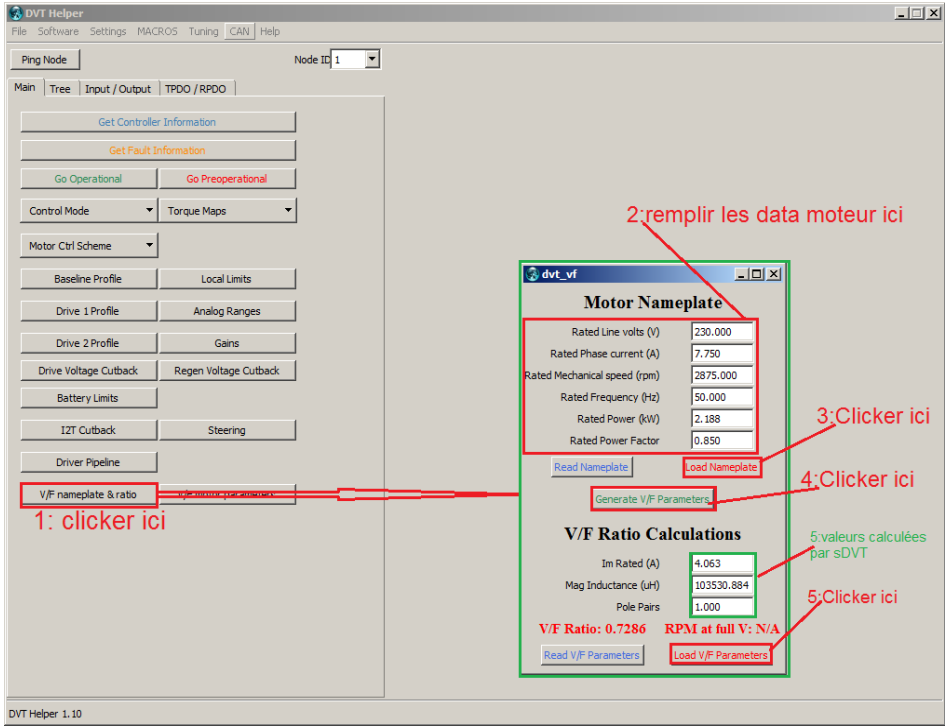
- Exemple Moteur LSES90L (câblage Triangle / delta)
  - Alimentation : 330V
- Variateur HVLP :
- Firmware version : SN0088.31 [SN0089.31]
- Configuration initiale = dcf pour le moteur LSES100L
- Mise à jour des paramètres pour le moteur LSES90L → suivre procédure suivante.

- Procedure de mise à jour de la configuration pour le moteur LSES90L.
- Dans DVT_helper
- Mettre le variateur en PreOp.
- Mise à jour des data moteur.

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V/F Control Setup



Note : les données correspondent aux valeurs pour le moteur **LSES90L câblé en Triangle**

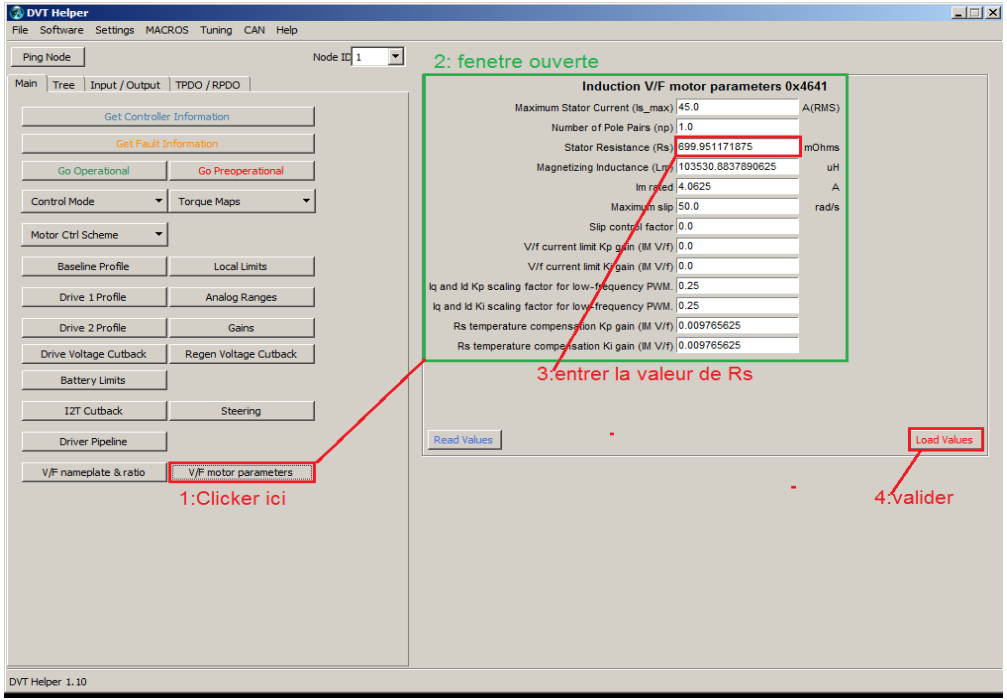
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V/F Control Setup

Mise à jour de la valeur de Rs :



Valeur de Rs :

Mesurer la résistance entre deux phases (à vide) et diviser par 2 !

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Thank you!



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